

Exploring Intervention Strategies for Distracted Students in VR Classrooms

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ABSTRACT

As one of the manifestations of virtual reality (VR) in education, virtual classroom allows students to enjoy a near-real classroom experience. VR class creates much better engagement and helps to stimulate interest and motivation in learning, making it an ideal solution to online teaching and learning activities, especially during the COVID-19 pandemic. Distraction is an unavoidable problem in immersive virtual classes, which has a great detrimental impact on learning. However, how to intervene in students' distraction behaviors in immersive virtual environments has not been thoroughly investigated up to now. In this paper, inspired by teachers' instructional techniques in real-life classes, we propose three intervention strategies, namely eye contact, verbal warning and text warning, and explore the intervening effects of these strategies on the inattention of students seated at the front or back of a virtual classroom via eye tracking. Our results show that all of the proposed intervention strategies have positive impacts on the attention of students. This research gives evidence that teachers' instructional techniques in the real world can be transferred to the virtual class, which provides a new insight for the future design of educational VR.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI.

KEYWORDS

distraction, attention, intervention, virtual reality

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1 INTRODUCTION

VR is a prospective technology and has attracted a lot of attention because of its enormous potential in revolutionizing education [1-3], which enables the possibility of situating students in a computer-generated context that would be too far, dangerous, or even nonexistent in real life [4-7]. As one of the presentation forms of educational VR, virtual classroom provides better engagement and allows students to obtain embodied experiences similar to real classes, which makes it an ideal solution to instructional activities compared with traditional online and remote learning through audio and video, particularly during the COVID-19 pandemic. However, there exist distractions in VR that would turn students' attention away from learning [8], which means that students may be attracted by other objects that are not relevant to the learning content during class.

Various studies from psychology, cognitive science and neuroscience have manifested that distraction has a great adverse impact on learning [9-16]. When students are distracted during learning, their brains actually process and store information in a different, less useful way, making them much less adept at extending and extrapolating to new contexts. Moreover, dividing students' attention during the encoding will impair their subsequent memory in relation to what they are working on [17]. To deal with the problem of distraction, teachers in real-world classes usually adopt instructional techniques such as making eye contact and giving verbal warnings to maintain discipline and enhance students' attention, which has already been proved effective by prior studies [18, 19].

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Nevertheless, the effectiveness of eye contact and verbal warning in virtual classes has not yet been thoroughly investigated.

Eye tracking can be exploited to study the relationships between eye movement data and cognitive activities [20], which is capable of reflecting learners' states of concentration. Eye tracking has long been investigated as an efficient tool to dissect learners' behaviors [21–23]. In recent years, consumer VR headsets with embedded eye trackers have become available, which open the way to gain a better insight into students' shifts of visual attention and interests in immersive VR. With the help of eye tracking, researchers are able to analyze and understand students' learning states more comprehensively under different VR settings.

In this paper, we explore different intervention strategies for distraction behaviors in VR classes via eye tracking. Inspired by teachers' intervention methods in real-world lectures and taking the unique advantages of VR into account, we propose three intervention strategies: eye contact, verbal warning and text warning, among which eye contact belongs to implicit ways and the other two are explicit ways. Additionally, the seat location, which means that students sit proximally to the virtual teacher or further away, is considered as another potential factor affecting attention. Thus, our study has two different design factors in total. The intervening effects of these strategies on the distraction behaviors of students seated in different positions are also evaluated. To the best of our knowledge, this is the first work that assesses intervention strategies for distracted students in an immersive VR classroom, which may contribute to the optimal design of educational VR environments.

2 RELATED WORK

As our work primarily centers on distraction behaviors in VR classes, we highlight three essential related bodies of work, VR in education, distraction in VR, and eye tracking in VR.

2.1 Virtual Reality in Education

Many studies have been conducted to explore the effectiveness of educational VR systems on teaching and learning. The impacts of virtual learning environments on fostering students' learning (academic achievements, motivation, engagement et al.) in education have been emphasized in [24–33]. Apart from the applications in students' learning, Stavroulia et al. [34] conducted a preliminary investigation into the usability of VR for teacher training and professional development. Kim et al. [35] proposed a VR learning management system that was regarded as a fully-featured virtual MOOC system. For maximum effectiveness, how teachers should perceive and design educational activities in immersive VR was studied in [36].

2.2 Distraction in Virtual Reality

Attention is never failed to be emphasized under any circumstance. People in immersive VR environments might be distracted due to mind-wandering, unwanted noise, external interference, etc. However, few studies have investigated the split-attention in VR. Voinescu et al. [37] studied the influence of different sensory modalities on attention in a VR aquarium and the result proved that the attention performance measured by the number of missing errors

was better under the visual condition. What's more, subtle attention guidance has been explored to navigate ways in immersive VR environments with the least possible distraction [38]. In educational situations, an interface to keep student indicators within the teacher's field of view was designed to help teachers notice the distracted ones [39]. Sitting positions [40, 41] and virtual peer-learners' behaviors [41] were also considered to be the factors that affect attention. Moreover, the deep learning approach was used to classify the distraction level of students in educational VR environments [42].

2.3 Eye Tracking in Virtual Reality

In recent years, VR headsets with embedded eye trackers have advanced rapidly, making it possible to further explore interaction techniques through natural eye movements in VR experiences. Previous research has investigated the classification of facial expressions [43] and the "hands free" interaction [44] in VR via eye trackers. Several other works focused on detecting users' attention via eye tracking in VR systems [45–47]. Besides, exploring users' behaviors in educational VR via their attention has been studied over the past few years. It was found that gaze tracking could detect whether students' focus was diverted away from the critical objects [48]. Gaze data of students was also available and visualized for teachers to measure their attention level [49, 50]. Furthermore, [41] and [51] investigated the effects of different VR classroom configurations on learners' visual attention and interaction through their eye movements respectively.

3 METHODOLOGY

In this study, a 4×2 factorial between-subjects design was adopted to compare and analyze the effects of different intervention strategies for distraction given two factors: intervention strategy and sitting position. The levels of intervention strategy were eye contact, verbal warning, text warning, and no intervention to serve as the control; the levels of sitting position were seating the participants either at the front or back of the virtual classroom.

3.1 Participants

A total of 120 participants comprised of undergraduate and graduate students from a university of China were recruited through social networks for the study. Nine participants were excluded from data analysis due to incorrect calibration or procedural issues, with the final sample consisting of 111 participants (52 male and 59 female), ages ranging from 18 to 30 ($M = 23.51$, $SD = 1.86$). Among the 111 participants, 50 (45.05%) had no previous experience with VR, 57 (51.35%) had some previous experience with VR and 4 (3.60%) declared themselves to be familiar with VR. None of them had any prior knowledge about the lecture content.

3.2 Apparatus

The experimental setup included an HTC Vive Pro Eye with integrated Tobii eye trackers, a 27-inch Philips monitor and a gaming PC (the main specifications are Intel Core i7-8700K @ 3.70 GHz, Nvidia GTX 1080 TI Graphics card, 32 GB RAM) with Microsoft Windows 10. Unity, a 3D real-time development platform published



Figure 1: Views from the virtual classroom: (a) overall design of virtual classroom; (b) virtual teacher with eye contact; (c) virtual classmates; (d) text warning for distraction.

by Unity Technologies, was employed for implementing VR techniques.

3.3 Experimental Design

To implement the experiment, a system named “Virtual Class” was developed. We choose a lecture theatre model to build the virtual classroom so that participants’ vision has less chance of being obscured by the virtual classmates in the front rows. The virtual classroom mainly consists of seven rows of desks, and each row has seven individual chairs. The virtual classroom also includes other objects such as rostrum, blackboard, projector and screen, which is the same as the real one. Twenty-six virtual classmates are arranged in the classroom, with each row seating three to four classmates. During lecture time, the topic is written on the blackboard and the slides are presented on the screen. The virtual teacher walks around the rostrum, similar to what a real teacher does. All of the virtual classmates keep sitting on chairs and listening carefully throughout the lecture. Figures 1 (a), (b), and (c) show the overall virtual classroom design, virtual teacher, virtual classmates, respectively. The content of the lecture is Von Neumann architecture and hardware composition of a computer. It should be noted that the learning topic can be replaced with any other content, especially those invisible or dangerous in the real life. This topic is selected because it is relevant to everyday life. Students will not get bored and are able to appear a relatively natural learning state. The total time of the lecture is about 10 minutes.

Our system contains four modes, namely eye-contact mode, verbal-warning mode, text-warning mode and non-intervention mode. During the lecture, participants will be judged as distracted by the system if they turn their eyes away from the teacher or screen for 2 seconds. Once detecting distraction, the system can apply different response strategies under different modes: in the eye-contact mode, participants can listen to the lecture freely, the teacher keeps looking at the participants throughout the lecture, no other intervention will be implemented for participants’ distraction behaviors, as shown in Figure 1 (b); in the verbal-warning mode, the teacher will pause the lecture and give a verbal warning to remind participants to pay attention to the lecture content; in the text-warning mode, the teacher will not pause the lecture but a red warning box will pop up in front of participants’ eyes and blink three times, as shown in Figure 1 (d); in the non-intervention mode, the system settings are almost the same as the eye-contact mode but without the virtual teacher’s eye contact. In addition, each mode contains two conditions, i.e. sitting at the front or back. Participants sitting at the front means that participants are seated in the first row, whereas participants sitting at the back are seated in the fourth row to ensure that they can get eye contact from the

virtual teacher. Participants under both conditions are located in the middle of the row. Consequently, our design leads to $4 \times 2 = 8$ treatment groups in total.

3.4 Procedure

Before arrival, participants were randomly assigned to one of eight treatment groups. Upon arrival, participants were first briefed about the content and the procedure of the experiment. To maintain natural behaviors, participants were blinded to the true purpose and design of the study, as it was solely introduced as an experience of a virtual class. Then, participants were asked to sign a consent form and fill out a demographic questionnaire. Afterwards, participants sat on chairs, wore the VR headsets with the help of experimenters, and familiarized themselves with the hardware and the VR environment. The experiment started with the calibration of eye trackers. After the eye trackers were calibrated successfully, the lecture and data collection started with a class ring. At the end of the virtual lecture, participants were told to take off the VR headsets by the virtual teacher. It took about 15 minutes to complete the entire experimental process per participant in total.

3.5 Data Collection

To evaluate the effects of intervention strategies on students’ level of concentration, the total amount of time spent on each of the objects-of-interests in the VR classroom, which was called visual attention in prior study [41], was collected and analyzed. We chose the virtual teacher, virtual classmates, screen, desks, chairs, walls, floor and ceiling as objects-of-interests (hereinafter referred to as objects), which were highly relevant to attention towards learning. In particular, we classified the aforementioned objects except the teacher and screen as other objects, since they were not relevant to the lecture content. It should be mentioned that classmates might be related to the learning content when classmates attend the lecture by hand-raising or verbal inputs. In this experiment, we defined classmates as objects irrelevant to the learning content, as they did not behave in any way other than sitting and listening throughout the lecture. During the experiment, the total amount of time that participants spent on the above-mentioned objects was calculated and recorded through head location and rotation as well as gaze direction data, which could be obtained via the software development kit of HTC Vive Pro Eye. In addition, as gaze data collected by the eye trackers can be affected negatively by blinks or noisy sensor measurements, we applied a linear interpolation on the gaze direction vectors to clean the data. In order to get rid of the interference that people gaze at some objects unconsciously for a very short time, we set an attention threshold of 200 ms based on prior work [41], which means that the gaze time of a certain

Table 1: Key results of visual attention from statistical analyses. (EC: Eye Contact; VW: Verbal Warning; TW: Text Warning; NI: No Intervention; CG: Control Group; IS: Intervention Strategy; SP: Sitting Position)

Visual Attention On	Experimental Condition										
	EC		VW		TW		NI (CG)		IS (Main)	SP (Main)	IS × SP (Interaction)
	Front (N=14)	Back (N=14)	Front (N=13)	Back (N=14)	Front (N=14)	Back (N=14)	Front (N=14)	Back (N=14)	<i>p</i> -value	<i>p</i> -value	<i>p</i> -value
Teacher	98.64 (27.85)	56.88 (32.13)	83.49 (34.55)	62.73 (26.38)	88.88 (35.27)	61.47 (18.64)	60.76 (28.30)	47.43 (38.68)	0.019*	< 0.001***	0.361
Screen	349.44 (37.19)	329.41 (64.57)	408.12 (40.37)	375.68 (39.92)	373.50 (45.51)	342.91 (60.40)	286.73 (39.10)	269.26 (52.26)	< 0.001***	0.007**	0.920
Other Objects	14.27 (9.42)	43.75 (19.35)	6.79 (3.54)	11.95 (5.15)	5.32 (2.13)	11.15 (4.76)	52.70 (8.79)	90.91 (15.83)	< 0.001***	< 0.001***	0.749

object will be recorded by the system if participants gaze at the object for at least 200 ms. Finally, we could examine the effects of intervention strategies on drawing participants' attention through the comparison of the total amount of time spent on the virtual teacher, screen and so on between different treatment groups.

3.6 Data Analysis

Data analysis was conducted using IBM SPSS Statistics 25. Descriptive statistics were calculated to describe the means and standard deviations, analysis of variance (ANOVA) and Scheirer-Ray-Hare test were adopted to identify the significant effects.

3.7 Research Questions

This study is aimed at answering the following two research questions:

- **RQ1:** Does students' level of concentration benefit from the intervention strategies for distraction in educational VR?
- **RQ2:** Whether teachers' instructional techniques in real-world classes can be transferred to immersive VR classes?

4 RESULTS

Within the scope of this study, students' level of concentration under different interventions was determined. Data are reported as statistically significant at $p < 0.05$. *, **, and *** correspond to the significance levels of $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively. Table 1 shows the key results from statistical analyses (the values in parentheses refer to the corresponding standard deviations).

4.1 Visual Attention on Teacher

Prior to statistical analyses, Shapiro-Wilk and Levene's tests were used to confirm that both of the assumptions of normality and homogeneity of variance were satisfied. As a result, a two-way ANOVA was conducted to evaluate whether participants' visual attention on teacher differed across the eight treatment groups. The dependent variable was the total time spent on the teacher, and the means and standard deviations for the time are presented in Table 1. The results indicated significant main effects for intervention strategy $F(3,103) = 3.447$, $p = 0.019$, partial $\eta^2 = 0.091$, and sitting position $F(1,103) = 19.538$, $p < 0.001$, partial $\eta^2 = 0.159$. However,

no significant interaction effect was found between intervention strategy and sitting position, $F(3,103) = 1.079$, $p = 0.361$. Subsequent Fisher's Least Significant Difference (LSD) post-hoc tests showed that attention time on the teacher of the control group was significantly shorter than that of the other three intervention groups, as shown in Figure 2 (a). The results also illustrated that the participants sitting at the front attended to the virtual teacher significantly more than those sitting at the back, as shown in Figure 2 (b).

4.2 Visual Attention on Screen

Before statistical analyses, the assumptions of normal distribution and homogeneity of variance were verified by the Shapiro-Wilk and Levene's test, respectively. Hence, parametric analyses were applied using two-way ANOVA and subsequent post-hoc LSD tests, where applicable. The total attention time on the screen was found to differ significantly between different intervention conditions with $F(3,103) = 26.842$, $p < 0.001$, partial $\eta^2 = 0.439$, as shown in Table 1. Simultaneously, the interaction between intervention strategy and sitting position was insignificant, $F(3,103) = 0.164$, $p = 0.920$. Post-hoc multiple comparisons showed that attention time on the screen of the control group was significantly shorter than that of the other three intervention groups, as shown in Figure 2 (c). The results also revealed that students in the verbal-warning condition spent significantly more time gazing at the screen compared with those in the eye-contact and text-warning conditions. Meanwhile, we found that the participants sitting at the front attended to the screen for a significantly longer period than the back seated ones with $F(1,103) = 7.449$, $p = 0.007$, partial $\eta^2 = 0.067$, as shown in Figure 2 (d).

4.3 Visual Attention on Others

A Shapiro-Wilk test was conducted and the results showed that the assumption of normality was not satisfied. A Scheirer-Ray-Hare test, which is a non-parametric test, was therefore applied to evaluate the effects of different intervention strategies on participants' visual attention on other objects. The results showed significant main effects for intervention strategy $H(3,103) = 72.601$, $p < 0.001$, $\eta^2 = 0.650$, and sitting position $H(1,103) = 15.442$, $p < 0.001$, $\eta^2 = 0.132$, as shown in Table 1. Besides, an interaction between intervention strategy and sitting position could not be demonstrated, $H(3,103) = 1.215$, $p = 0.301$.

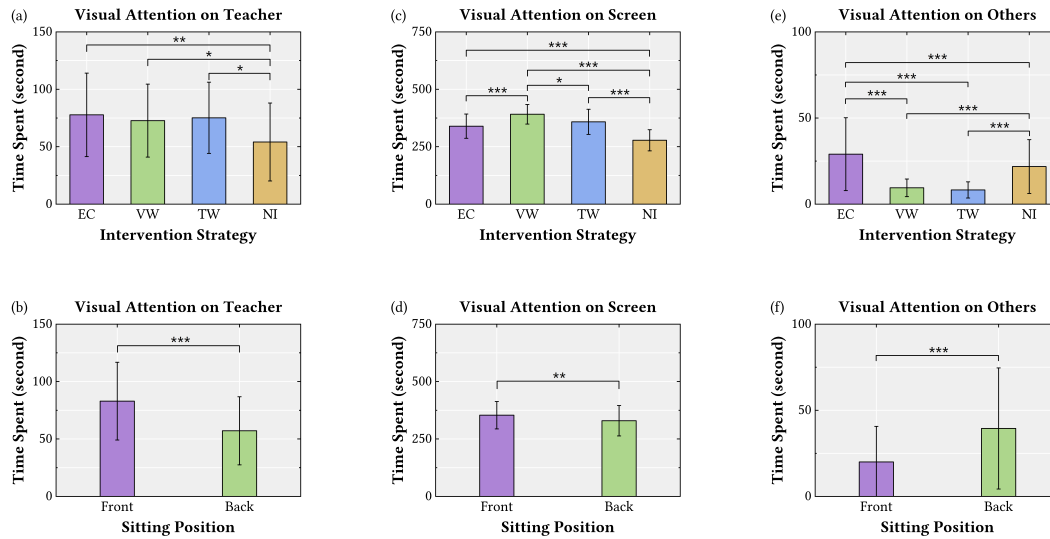


Figure 2: Results of visual attention: (a) comparison of visual attention on teacher between intervention strategies; (b) comparison of visual attention on teacher between sitting positions; (c) comparison of visual attention on screen between intervention strategies; (d) comparison of visual attention on screen between sitting positions; (e) comparison of visual attention on other objects between intervention strategies; (f) comparison of visual attention on other objects between sitting positions. (EC: Eye Contact; VW: Verbal Warning; TW: Text Warning; NI: No Intervention)

= 0.749. Follow-up post-hoc Kruskal-Wallis H tests were conducted to evaluate pairwise differences between means across the levels of intervention strategy. The results demonstrated that students from the eye-contact and control groups spent significantly more time on other objects than those from the other two groups, as shown in Figure 2 (e). In particular, the total time spent on other objects in the control and eye-contact groups differed significantly from each other. And the results of the Mann-Whitney U test suggested that participants sitting at the back spent more time on other objects, as shown in Figure 2 (f).

5 DISCUSSION

In this section, we mainly discuss the intervention for distraction, the privacy preservation, the guidelines for the design of educational VR as well as the limitations of this study and implications for future work.

5.1 Intervention for Distraction in Virtual Reality Class

Results related to the intervention for distraction in the virtual class are particularly discussed in two parts, the impacts on maintaining concentration (RQ1) and the practicability of transferring teachers' instructional techniques in real-world classes to immersive VR (RQ2).

In relation to RQ1, we argue that the proposed intervention strategies for distraction would benefit students' level of concentration. We found that students generally did concentrate more on the learning content and less on other objects in the intervention groups compared with those in the control group, both in terms of sitting at the front and back of the classroom. Such is human

nature, in case people are reminded when they are distracted, they will undoubtedly pay more attention to what they are supposed to concentrate on. Additionally, there was no significant statistical difference across the three intervention strategies in the results of the attention time on the teacher. We infer that the degree of the influence of these strategies on students' attention time varies slightly. Besides, students in the verbal-warning condition paid more attention to the screen and less to other objects than those in the eye-contact condition. A likely explanation for this result may be that the explicit intervention is more effective than the implicit one to some extent. Another interesting finding comes from the comparison that students in the verbal-warning condition spent more time on the screen than those in the text-warning condition. This difference may be attributed to the fact that teachers usually give verbal warnings in real classes and students are more sensitive to teachers' verbal warnings. Moreover, students seated in the front apparently spent more time on the teacher and screen and less time on other objects than those seated in the back, which is in line with previous research [41].

Furthermore, this study also focuses on whether teachers' instructional techniques in real-life classes can be transferred to immersive VR classes (RQ2). In this regard, the intervention strategies of eye contact and verbal warning come from the instructional techniques of real teachers, which have been validated in real classes by prior works [18, 19]. Our results manifested that the students from the eye-contact and verbal-warning groups had a higher level of concentration compared with those from the control group, proving that these instructional techniques are also effective in VR. Hence, our findings provide support for the feasibility and validity of transferring teachers' instructional techniques in real-world classes to

the educational VR, which gives a novel idea for the future design of teacher-student interaction in virtual classrooms.

5.2 Privacy Preservation

Student privacy is an important issue when collecting eye-gaze data during the learning process. In our study, we address privacy challenges by only collecting processed data (i.e. the total amount of time spent on the objects) from participants who gave permission to use their data within a standard informed consent model, and without recording any information that could be used to fully de-anonymize it later. The goal of this study is to determine the effectiveness of the proposed intervention strategies rather than to better monitor the students' behaviors in educational VR. In the design of an educational VR system, these intervention strategies should be packed as an alternative assistant tool to help students concentrate on the learning content. The system must not record user data and users are quite within their rights to refuse to be intervened.

5.3 Design Considerations

In addition to providing empirical evidence verifying the effectiveness of the intervention strategies for distraction in an immersive educational context, our study revealed a range of findings that designers and researchers could consider when designing VR classroom applications. With the goal of promoting students' learning performance in VR classes, here we attempt at generalizing the most important findings to provide the following applicable design suggestions:

- Eye contact of the virtual teacher is recommended in priority to gain students' attention.
- Verbal warning and text warning can be adopted if students distract frequently.
- Students should be allocated to seats close to the virtual teacher.
- The design of teacher-student interaction in educational VR might as well start with teachers' instructional techniques in real-life classes.

5.4 Limitations and Implications for Future Work

There are still several limitations regarding the present study that should be noted. Firstly, the accuracy of eye trackers can be easily affected by the cleanliness of lenses for people who wear glasses, which might further influence the precision of the experimental results. Secondly, eye trackers cannot identify such distraction behaviors as students are looking at the learning content but are still distracted mentally, which can result in the invalidity of our intervention strategies. A possible way to overcome this limitation is to collect electroencephalography (EEG) signals as an assistant measure in future studies. Thirdly, it cannot be denied that experiences involving VR headsets remain quite novel in educational contexts. As a majority of the participants had little experience with VR devices, we cannot rule out whether the novelty of VR caused distraction during the lecture. Lastly, it is essential to point out that this study includes a limited sample size, and a larger,

multi-centric study must be designed and carried out to draw more credible conclusions.

Our research proposed basic solutions to keep students' attention in immersive virtual learning environments. In order to better understand students' learning behaviors, however, the impacts of these intervention strategies on the academic achievements and level of flow state need to be further studied. A future study could look at investigating the relationship between students' academic achievements, level of flow, and visual attention.

6 CONCLUSION

To explore solutions to distraction behaviors in VR classrooms, this study proposed three intervention strategies and evaluated students' visual attention under different interventions based on eye tracking. We found that all of the proposed intervention strategies are effective approaches to arouse students' attention. Participants in the intervention groups focused more on the virtual teacher and screen, while participants in the control group spent more time on the objects irrelevant to the learning contents. In addition, participants sitting at the back were more easily to be attracted by the objects which were not relative to the lecture compared with those sitting at the front of the virtual classroom. Our findings give evidence that teachers' instructional techniques in the real world can be transferred to the virtual class, which provides a new way to the future design of educational VR.

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